Lesson plan · for the teacher

Data and automation: unit check

IoT · Level 3 · Unit check

Time
45 min

Materials
Board, chalk, paper. No devices needed.

Learning goals

- Logging data you can trust
- IF/THEN: rules that act
- Power: batteries and the sun
- Networks: getting the data home

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- What will your pupils build or measure first?

Exit ticket

- One idea I will use, and when I will use it

Try it in class tomorrow
Pick the idea from this unit your pupils need most, teach it this week, and tell a colleague how it went.

Worksheet · for pupils

Data and automation: unit check

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Name _____ Class _____ Date _____

1 **Before trusting a new classroom temperature sensor, what should you do first?**

- ◯ Check that its chart over a day looks smooth
- ◯ Compare it with a trusted thermometer for an hour
- ◯ Average a week of its readings to even out errors

2 **Six readings, sorted: 26, 27, 29, 30, 31 and 88 °C.**

The median is _____ °C, and the average is _____ °C.

Word bank: 29.5 · 38.5 · 29 · 88

3 **With two thresholds (fan on above 30 °C, off below 28 °C), a room at 29 °C always has the fan off.**

◯ True ◯ False

4 **A tank's level sensor may fail. Which designs fail safe?**

- ☐ Stop the pump if readings stop arriving
- ☐ Keep pumping until someone sees the overflow
- ☐ Stop the pump if a reading is impossible, such as -5 %
- ☐ Alert the caretaker whenever the pump is stopped this way
- ☐ Run the pump on a timer instead

5 **A sensor averages 0.5 mA on two AA batteries (2,500 mAh). Planning with 70 %, about how long will it last?**

◯ About 3 weeks ◯ About 5 months ◯ About 7 months